Critical inverse moments and local observation geometry on finite flat lattices

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Abstract

This appendix consolidates one sufficient coefficient condition and its limitations in the existing flat-lattice observation-rank dossier. A global inverse-conductance moment of order $D/2$ suffices for the mean-budget rank order through an actual real spectral observation. Every lower positive inverse exponent admits explicit divergence-free counterexamples. At the critical exponent, spatial observations by means on a full dyadic cube tree can require logarithmically many blocks even for a fixed weak region; the same profiles admit a constant-size unrestricted rectangular partition. The finite Fourier and flow adapters are proved analytically, and exact finite diagnostics are recorded separately from formal verification. A four-cycle counterexample now has a focal Lean proof: coercivity on the zero-mean kernel does not justify the same fixed penalty. The distinction from a freely chosen penalty is essential. We also give a general rectangular bound with exponent $D/4$, an obstruction to its single-threshold mechanism, and the target exponent for supplied scaled cubes. A fixed-moment two-neighbour example prevents deriving that exact cube decomposition universally. The spectral and geometric inputs are classical; independent novelty, an interacting gauge-theory gap and a continuum limit are not established. This is a consolidation appendix, not a separate paper or AIRR deposit.

1 Actual objects and the original observation problem

Fix $D \geq 3$, $N_c \geq 2$, $c > 0$, and the unit-spacing torus $X_n = (\mathbb{Z}/n\mathbb{Z})^D$. The actual real adjoint coefficient space has dimension $g = N_c^2 - 1$; there are D field components, so write $h = Dg$. For the actual forward difference $d_i u(x) = u(x + e_i) - u(x)$, take one scalar total coefficient $a(x) = 1 + w_{i,x,j} > 0$ shared by all derivative directions and field components. Use counting norms and the actual componentwise flat energy

$$Q_w(U) = \mathcal{E}_a(U) = \sum_{x,i,j} a(x)\|d_i U_j(x)\|^2. \tag{1}$$

The existing flat Hodge identity identifies these forms; its formal source is listed in Section 7. All new adapters in this appendix remain analytic. There is no substitution of a covariant or interacting derivative in (1).

The original local data consist of a differentiable map C at zero, with $C(0) = 0$, into an actual finite-dimensional real normed space F, and a penalty satisfying $\Phi(y) \leq K\|y\|^2$ near zero, such that

$$c\|U\|^2 \leq Q_w(U) + \Phi(C(U)) \tag{2}$$

for sufficiently small actual divergence-zero fields. "Local" here means a configuration-space neighbourhood; spatial support of C is an additional condition. The constant c is fixed, while observation maps and neighbourhoods may depend on n.

A useful necessary condition follows directly. If an actual J-dimensional divergence-zero subspace has $Q_w(U) \le \lambda\|U\|^2$ with $\lambda < c$, then $\dim F \ge J$. Otherwise a nonzero U in that subspace lies in $\ker DC(0)$. Along tU, $C(tU) = o(t)$ and the penalty is at most $o(t^2)$. Dividing (2) by t^2 and taking $t \to 0$ contradicts $\lambda < c$. No rank or band certificate is supplied as an extra hypothesis in this argument.

2 The critical global moment theorem

Put $p = D/2 > 1$. For $m \ge 1$ suppose that

$$n^{-D} \sum_{x \in X_n} (m/a(x))^p \le M < \infty. \tag{3}$$

Theorem 1. *Under* (3)*, there is an actual real linear spectral observation C such that*

$$c\|U\|^2 \le Q_w(U) + c\|C(U)\|^2 \quad \textit{for every actual cochain } U,$$

and

$$\dim F \le C_* \max\{1, (n/\sqrt{1+m})^D\}, \tag{4}$$

where, with $I_p = (p-1)^2/[p(p+1)]$ and $\Gamma_p = \frac{p}{p-1} I_p^{-(p-1)}$, one safe choice is

$$C_* = h\left[1 + \Gamma_p (3/4)^D D^{p+1} c^p M 2^p\right].$$

Thus $\Phi(y) = c\|y\|^2$, $K = c$ and any positive neighbourhood radii supply the original local data. In particular the assertion holds on the actual divergence-zero subspace.

The original subclass $1 \le a \le 2m$ with exact arithmetic mean m is included, but those extra normalizations are not needed for sufficiency. The distribution condition (3) remains additional; it is not derived from an arithmetic mean.

2.1 A finite normalized Fourier lemma

For a finite abelian group of size V, let $\mathcal{F}$ be its counting-unitary Fourier matrix, whose entries have modulus $V^{-1/2}$. For a frequency multiplier g put

$$L = V^{-1} \sup_{t>0} t^p \#\{k : |g(k)|^2 > t\}.$$

For a nonnegative matrix B, write $\mathsf{n}(s, B)$ for its number of eigenvalues strictly greater than s, counted with ordinary multiplicity.

Lemma 1. *For $K = M_f \mathcal{F}^* M_g \mathcal{F}$ and $s > 0$,*

$$\mathsf{n}(s, K^*K) \le \Gamma_p L s^{-p} \sum_x |f(x)|^{2p}.$$

The trace and eigenvalue counts are not volume-normalized.

Proof. Remove phases by unitaries and set $A = \mathcal{F}^* M_{|g|^2} \mathcal{F}$. If $L = 0$ the assertion is immediate. Otherwise let $B_0 = pL/(p-1)$ and let A^+ be the inverse on $\operatorname{ran} A$, zero on its kernel. For $0 \le \gamma \le A$ write $\rho(x) = \langle \delta_x, \gamma \delta_x \rangle$. Positivity makes γ annihilate $\ker A$. On the range let $P_E = \mathbf{1}_{\{A^+ \le E\}}$ and $R_E = \mathbf{1}_{\{A^+ > E\}}$. The weak-frequency tail gives

$$\langle \delta_x, P_E A P_E \delta_x \rangle = V^{-1} \sum_{|g(k)|^2 \ge 1/E} |g(k)|^2 \le B_0 E^{p-1}.$$

Indeed the tail above t is t times its count plus the integral of counts above t; the resulting bound is $VLp\, t^{1-p}/(p-1)$. Non-strict thresholds follow by limits. The triangle inequality for $\gamma^{1/2}\delta_x$ split by $P_E + R_E$ yields

$$\langle \delta_x, R_E \gamma R_E \delta_x \rangle \ge \big(\sqrt{\rho(x)} - \sqrt{B_0} E^{(p-1)/2}\big)_+^2.$$

Exact spectral layer integration consequently gives

$$\operatorname{Tr}(A^+ \gamma) \ge I_p B_0^{-1/(p-1)} \sum_x \rho(x)^{p/(p-1)}. \tag{5}$$

The scalar integral reduces to $\int_0^1 (1 - t^{(p-1)/2})^2 dt = I_p$. Now take P to be the spectral projection of $A^{1/2} M_{|f|^2} A^{1/2}$ above s, with rank $\nu > 0$, and set $\gamma = A^{1/2} P A^{1/2}$. Then $0 \le \gamma \le A$, $\operatorname{Tr}(A^+\gamma) \le \nu$, and

$$s\nu < \sum_x |f(x)|^2 \rho(x) \le \Big(\sum_x |f(x)|^{2p}\Big)^{1/p} \Big(\sum_x \rho(x)^{p/(p-1)}\Big)^{(p-1)/p}.$$

Insert (5) and raise to p. The constant becomes $B_0 I_p^{-(p-1)} = \Gamma_p L$. This also explicitly handles a zero-frequency kernel. □

The density method is classical and follows the Rumin–Frank approach [3, Section 2]. The proof above supplies its finite normalization and kernel adapter; it is not a claim of a new Cwikel mechanism or independent discovery.

2.2 Actual lattice frequencies and the weighted inverse

For $n \ge 2$ let $L_0 = d^* d$ and work first on the actual counting-mean-zero scalar space H_0. Its Fourier symbols are

$$z_i(k) = e^{2\pi i k_i/n} - 1, \qquad \lambda(k) = 4 \sum_i \sin^2(\pi k_i/n).$$

Choose centred integer representatives r_i with $|r_i| \le n/2$, using only one representative at an even endpoint. Concavity of sine implies

$$\lambda(k) \ge 16 |r|^2/n^2, \qquad |g_i(k)| := |z_i(k)/\lambda(k)| \le n/(4|r|), \quad k \ne 0.$$

Set $g_i(0) = 0$. If $|g_i| > \tau$, then $0 < |r| < R = n/(4\tau)$. For $R \le 1$ there are none, and for $R > 1$ there are at most $(3R)^D$ representatives. Thus

$$n^{-D} \sup_{t>0} t^{D/2} \#\{|g_i|^2 > t\} \le (3/4)^D. \tag{6}$$

This is uniform in n, including $n = 2$, not an imported infinite-volume bound.

Let $L_a = d^* M_a d$. On H_0 it is positive definite. For $f \in H_0$, the flow $j_* = M_a d L_a^{-1} f$ minimizes $\langle j, M_{a^{-1}} j \rangle$ among $d^* j = f$: write $j = j_* + k$, $d^* k = 0$, and the cross term is $\langle d L_a^{-1} f, k \rangle = 0$. Inserting the explicit feasible flow $j = d L_0^{-1} f$ gives the actual quadratic-form comparison

$$L_a^{-1} \le \sum_{i=1}^{D} T_i^* T_i, \qquad T_i = M_{a^{-1/2}} d_i L_0^{-1}. \tag{7}$$

Extend L_0^{-1} by zero on constants. Lemma 1 and (6) bound each $\mathsf{n}(s, T_i^* T_i)$ by

$$\Gamma_p (3/4)^D s^{-p} \sum_x a(x)^{-p}.$$

For nonnegative matrices, the count of their sum above t is no larger than the sum of their counts above t/D: intersect the complements of their high spectral subspaces and apply min–max. No commutation is required. With $t = 1/c$ in (7), inversion on H_0 and the single scalar constant mode yield

$$\#\{\lambda(L_a) < c\} \le 1 + \Gamma_p (3/4)^D D^{p+1} c^p M n^D / m^p. \tag{8}$$

No spectral-band dimension is assumed in this derivation.

Completion of Theorem 1. Choose an actual real counting-orthonormal eigenbasis of the real symmetric matrix L_a. Retain the coordinates with eigenvalue $< c$, separately in each of the h real field coordinates. Complexification used in the Fourier proof preserves real eigenvalue multiplicities; it does not double them. Energy controls coefficients at eigenvalues $\ge c$, while $c\|C(U)\|^2$ controls the rest. Equation (8), $p = D/2$, and $(1+m)/m \le 2$ give (4). For $n = 1$ take the identity on all h coordinates. This constructs the actual global spectral observation. □

3 Every lower positive inverse exponent is insufficient

Theorem 2. *For every fixed $0 < r < D/2$, the analogue of Theorem 1 under $\operatorname{mean}((m/a)^r) \le 2$ is false uniformly in the actual scalar class $1 \le a \le 2m$, $\operatorname{mean}(a) = m$.*

Proof. Choose a fixed integer $\ell > 3D/c$, and put $P = 2\ell + 2$, $B = (2\ell+1)^D$. The triangular window $t(s) = (1 - |s|/\ell)_+$ has finite sums

$$T = \sum t^2 = (2\ell^2 + 1)/(3\ell), \quad V = \sum (d^- t)^2 = 2/\ell, \quad W = \sum (d^+ d^- t)^2 = 6/\ell^2.$$

For a product window $\psi_y(x) = \prod_i t(x_i - y_i)$ and nonzero actual adjoint H, the actual curl $U_1 = d_2^- \psi_y H$, $U_2 = -d_1^- \psi_y H$, other components zero, has backward divergence zero. Its field support and all nonzero forward-gradient start sites lie in the box $y + [-\ell, \ell]^D$. Direct separable sums give

$$\|U\|^2 = 2VT^{D-1}\|H\|^2, \qquad \frac{\mathcal{E}_0(U)}{\|U\|^2} = \frac{W}{V} + (D-1)\frac{V}{T} = \frac{3}{\ell} + \frac{6(D-1)}{2\ell^2+1} \le \frac{3D}{\ell} < c.$$

Take sufficiently large multiples n of P, set $m = n^2$ and $J = \lfloor n^{D-2r}/B \rfloor$. These J boxes fit on the separated P-grid eventually, since $P^D \le B n^{2r}$. They satisfy $J \ge n^{D-2r}/(2B) \to \infty$. Put $s = BJ/n^D \le m^{-r}$ and set $a = 1$ on their union, $a = A = (m-s)/(1-s)$ elsewhere. Eventually $s \le 1/2$, so

$$\operatorname{mean}(a) = m, \quad 1 \le a \le 2m, \quad A \ge m, \qquad \operatorname{mean}((m/a)^r) \le s m^r + 1 \le 2.$$

Each curl's gradients start only in its weak box. The actual J-dimensional span has that same Rayleigh quotient, because both field and gradient supports are disjoint. Section 1 forces $\dim F \geq J$, whereas the target maximum at $m = n^2$ is 1. This proves the refutation using original nonlinear data. □

Together the two theorems give the sharp inverse-moment exponent for this specified family of global scalar conditions. They do not identify all sufficient coefficient criteria, prove a physical coefficient hypothesis, or impose spatial locality on C.

4 Spatial means and the critical dyadic obstruction

For a partition $\mathcal{P}$ into integer rectangles, its actual normalized means are

$$C_{\mathcal{P}}(U)_{R,j} = \sqrt{|R|}\,\mathrm{mean}_R\, U_j, \qquad \dim F = h\#\mathcal{P}.$$

A full dyadic cube tree splits each selected cube into all 2^D equal children and retains unsplit siblings. The origin of the torus coordinate cube may be translated. This class excludes arbitrary anisotropic rectangle cuts and compressed observations.

Proposition 1. *There are fixed critical-moment profiles with target maximum* 1 *for which every coercive full dyadic cube-tree mean partition has* $\Omega(\log n)$ *leaves, even after choosing a translated origin. The same profiles admit an explicit constant-size rectangular mean partition satisfying the global inequality for every actual cochain.*

Proof. Keep $\ell > 3D/c$ fixed. Choose a dyadic $L = 2^q \geq 4\ell + 2$, put $\gamma = \max(1, c/4)$, $n = 2^J$ with $J \geq q + 1$, $m = \gamma n^2$, and $S = \{0, \ldots, L-1\}^D$. Set $a = 1$ on S and $a = A = (m - s)/(1 - s)$ elsewhere, $s = L^D/n^D$. Then $1 \leq a \leq 2m$, its mean is exactly m, and

$$\mathrm{mean}((m/a)^{D/2}) \leq 1 + L^D\gamma^{D/2} =: M_0.$$

All these parameters except n are fixed, and the target maximum is 1.

In the hypothetical complete level $j_* = J - q$ of any translated dyadic grid, cells have side L. Each coordinate interval of S meets at most two such cyclic grid intervals, so one intersection has integer length at least $L/2 \geq 2\ell + 1$. Taking these intersections independently places a whole gradient-start box for a compact curl inside S and inside one level-j_* cell. If the proposed partition had maximum depth at most j_*, that cell would lie in one actual leaf. The curl then has zero means on every leaf, since each component has total sum zero, yet its weighted Rayleigh quotient is $< c$. This contradicts coercivity. A single zero-extended window needs no packing divisibility assumption on n; $n \geq 2L$ suffices here. The field may depend on the partition, since coercivity is asserted for every field. Thus maximum depth is at least $J - q + 1$, and the full-arity tree identity gives

$$\#\mathcal{P} \geq 1 + (2^D - 1)(J - q + 1). \tag{9}$$

For a successful unrestricted partition, isolate all L^D vertices of S as singletons, and use the D disjoint complement rectangles

$$R_i = [0, L-1]^{i-1} \times [L, n-1] \times [0, n-1]^{D-i}.$$

Coverage follows from the unique first coordinate outside $[0, L-1]$. Every internal edge starts at a strong vertex with coefficient A. On a length-l interval, counting variance and path Cauchy–Schwarz give

$$\sum \|u - \bar{u}\|^2 \le \frac{l(l-1)}{4A} \sum A\|du\|^2.$$

Indeed the path resistance is at most $(l-1)/A$, and a cut at t is crossed by $t(l-t) \le l^2/4$ pairs in the variance identity. Product-variance tensorization on original fibres yields a rectangle constant $n^2/(4A)$, with no extra factor D. Singletons have zero residual, internal edge sets are disjoint, and omitted edges have positive energy. Since $cn^2/(4A) \le c/(4\gamma) \le 1$, exact orthogonal decomposition proves

$$c\|U\|^2 \le Q_w(U) + c\|C_{\mathcal{P}}(U)\|^2, \qquad \#\mathcal{P} = L^D + D.$$

These rectangles have potentially unbounded aspect ratio; they leave the cube-tree class. □

The general critical moment question for unrestricted rectangular means remains open. Changing only the stopping rule within full cube trees cannot resolve it. Classical weighted Sobolev inequalities [4] also yield block observations for inverse moments of finite order strictly greater than $D/2$; the dossier contains the explicit balanced-box proof. Proposition 1 explains why its critical tree-counting loss cannot be removed merely by choosing a better cube-tree stopping criterion.

5 What a spatial observation proof must still establish

The preceding positive spectral result does not construct a rectangular mean map. Three further checks prevent substitutions for that missing construction. They concern local geometry, the penalty coefficient, and support of observations, respectively. They are collected here as limitations of the same dossier, not as additional papers.

5.1 Critical inverse mass does not remove rectangle geometry

For a rectangle R of side lengths l_i, set $V = \prod_i l_i$, $L = \max_i l_i$, $\kappa = L^2/V^{2/D}$ and $\mu = \sum_R a^{-D/2}$. The endpoint Sobolev estimate on balanced cubes, with $q = 2D/(D+2)$, transfers to arbitrary rectangles by repeated coordinate reflection and gives

$$\sum_R |u - \bar{u}_R|^2 \le A_D D^{2/D} \kappa \mu^{2/D} \mathcal{E}_{a,R}(u). \tag{10}$$

Here the energy uses original internal edges and their actual starting-vertex weights. The full reflection multiplicities and graph adapters to the classical estimate [4, Section 2.2] are preserved in the companion geometric note; this is an analytic adapter, not a new Lean theorem.

The factor cannot be discarded even for uniform coefficients. With $u(x) = x_i - (L-1)/2$ along a longest side, direct sums give

$$\sum_R u^2 = V(L^2-1)/12, \qquad \mathcal{E}_{a,R}(u) = aV(L-1)/L.$$

Their ratio is $L(L+1)/(12a) \ge \kappa\mu^{2/D}/12$. On the thin rectangle of lengths $(L, 1, \dots, 1)$, choosing $a = L^{2/D}$ makes $\mu = 1$ while this ratio diverges. Thus compressing a cube-tree chain into thin rectangles cannot simply reuse a shape-independent inverse-mass certificate. This prescribed-block test does not defeat every global partition: omitted crossing edges and a different partition may help. The successful partition in Proposition 1 is not contradicted.

5.2 Fixed and free penalties are different obligations

Let C be normalized means on a fixed partition of a finite real Euclidean space. Then $CC^* = I$; $P = C^*C$ is the orthogonal block-mean projection and $Q = I - P$. For the actual positive stiffness L_a, write the blocks on $\operatorname{ran} P \oplus \operatorname{ran} Q$ as

$$A = PL_aP, \qquad B = PL_aQ, \qquad D_0 = QL_aQ.$$

Positivity implies $\ker A \subseteq \ker B^*$: otherwise the energy of $tp + v$ would have a nonzero linear coefficient with zero quadratic coefficient in t. Completing the square with the finite spectral pseudoinverse $A^\dagger$ shows

$$L_a + cP \geq cI \quad \iff \quad D_0 - B^*A^\dagger B \geq cI \text{ on } \operatorname{ran} Q. \tag{11}$$

Indeed the first inequality is exactly $L_a - cQ \geq 0$, and its quadratic form is $\langle p + A^\dagger Bv, A(p + A^\dagger Bv)\rangle + \langle v, (D_0 - cI - B^*A^\dagger B)v\rangle$. This is classical shorted-operator algebra [2]; it supplies no spatial partition or target count. In particular $D_0 \geq cI$ alone is insufficient.

Proposition 2. *On the four-cycle with edge weights* $(1, 6, 4, 1)$ *and blocks* $\{0, 1\}, \{2, 3\}$*, coercivity holds on zero block sums with* $c = 19/5$*, but* $L_a + cP \geq cI$ *fails.*

Proof. For $v = (x, -x, y, -y)$, with the counting Euclidean norm,

$$\mathcal{E}_a(v) - \frac{19}{5}\|v\|^2 = \frac{17}{5}\left(x + \frac{35}{17}y\right)^2 + \frac{84}{85}y^2 \geq 0.$$

The actual normalized observation is

$$Cu = \left((u_0 + u_1)/\sqrt{2},\ (u_2 + u_3)/\sqrt{2}\right).$$

For $u = (19, -9, -5, -5)/14$, direct exact evaluation gives

$$\mathcal{E}_a(u) = 52/7, \quad \|u\|^2 = 123/49, \quad \|Cu\|^2 = 25/49,$$

and hence $\mathcal{E}_a(u) + (19/5)\|Cu\|^2 - (19/5)\|u\|^2 = -6/35$. The weights have mean 3 and lie between 1 and 6. Ten focal Lean theorems preserve the constructed weights, normalized observations, projection identities, kernel square identity and exact negative witness; Section 7 identifies the successful immutable source. This is a finite four-cycle result, not a proved embedding into the full physical divergence-zero kernel. □

For the original class allowing an arbitrary finite K, put $J = D_0 - cI$. The necessary and sufficient condition is instead

$$\exists K \geq 0 :\ L_a + KP \geq cI \quad \iff \quad J \geq 0 \text{ and } B \ker J = 0. \tag{12}$$

Necessity follows by testing $v \in \ker C$, then $v + tp$ with $v \in \ker J$. For sufficiency, $B \ker J = 0$ implies $B^*p \in \operatorname{ran} J$. Completing the square in $v + J^\dagger B^*p$ leaves $A + (K - c)I - BJ^\dagger B^*$, so $K = c + \|BJ^\dagger B^*\|$ suffices. In particular a proved all-cochain margin $D_0 \geq (c + \epsilon)I$ gives $K = c + \|L_a\|^2/\epsilon$. For actual $a \leq 2m$, the elementary edge estimate $|r - s|^2 \leq 2|r|^2 + 2|s|^2$ gives $\|L_a\| \leq 8Dm$, including the parallel edge occurrences at $n = 2$. A margin $2c$ therefore permits $K = c + 64D^2m^2/c$, without increasing the number of observations. No margin or partition is proved by this conditional implication. Ordinary block averaging need not preserve divergence, so

an estimate only on the physical kernel cannot silently replace the all-cochain hypothesis in this decomposition.

Even free K needs care at equality. On the same cycle let $s = \sqrt{85}$, $c_* = (17 - s)/2$ and $v = (7, -7, 6 - s, s - 6)$. Its kernel defect at c_* is zero, since for general $(x, -x, y, -y)$ it equals $(s - 6)[x + 7y/(s - 6)]^2$. For $p = (\delta/2, \delta/2, -\delta/2, -\delta/2)$ the full defect is

$$\mathcal{E}_a(v + p) + K\|C(v + p)\|^2 - c_*\|v + p\|^2 = (7 + K - c_*)\delta^2 - 10(13 - s)\delta.$$

For each finite $K \geq 0$ a sufficiently small positive δ makes it negative. This additional equality example is exact analytic algebra and finite diagnostic evidence, not one of the ten Lean theorems of Proposition 2.

5.3 Sparse point samples cannot replace stable averages

For uniform $a = m$, if an observation depends only on vertex values on a set S, $|S| = s$, and $C(0) = 0$, then the scalar field $u = 1$ off S and 0 on S has zero observation. At most $2Ds$ forward-edge occurrences cross S, so $\mathcal{E}_m(u) \leq 2Dms$. Consequently all-cochain coercivity requires

$$s \geq \frac{cn^D}{2Dm + c}.$$

For an actual divergence-zero test, fix a component direction j and let the field be constant along every j-line, equal to zero on each line meeting S and one on the other lines, in one fixed internal coordinate. There are at most s hit lines. Its backward divergence is zero, its norm squared is at least $n(n^{D-1} - s)$, and transverse crossing energy is at most $2(D - 1)mns$. Thus necessarily

$$s \geq \frac{cn^{D-1}}{2(D - 1)m + c}.$$

If every transverse line is hit, the second bound is immediate without a nonzero test. These bounds give the required obstruction: at $a = m = n^2$ and critical moment $M = 1$, bounded samples fail for all cochains when $D \geq 3$ and for these physical tests when $D \geq 4$. For the same uniform coefficients one global mean succeeds for $0 < c \leq 16$, by the nonzero Fourier gap $n^2\lambda(k) \geq 16$. This comparison rules out a point-sampling shortcut, not rectangular averages or singletons supplemented by complementary block means. A nonlinear local observation is tested on sufficiently small multiples of these fields.

6 Constructive rectangular bounds and their limits

Retain $m \geq 1$, $1 \leq a \leq 2m$, exact mean m, (3), and $p = D/2$. All assertions in this section remain analytic.

6.1 A general threshold construction

For an interval of length l, the identity $\mathrm{Var}(u) = l^{-1} \sum_{x<y} |u(x) - u(y)|^2$ and path Cauchy–Schwarz give coefficient at most $l(l - 1)/4$ per edge: each crossing pair has distance at most $l - 1$, and at most $l^2/4$ pairs cross any edge. Product variance tensorization therefore yields, for a rectangle of maximal side L,

$$\sum_R |u - \mathrm{mean}_R\, u|^2 \leq \frac{L(L - 1)}{4} \sum_{\text{internal forward edges of } R} |du|^2. \tag{13}$$

There is no additional factor D in this tensorization.

Fix $\tau \geq 1$. If $\tau < c/4$ use singletons. Otherwise set $\ell = \min\{n, \lfloor 2\sqrt{\tau/c} \rfloor\}$ and start with a grid whose sides have length at most ℓ. Isolate each vertex with $a < \tau$ as a singleton. Its current rectangle minus that vertex splits into at most $2D$ rectangles by the first coordinate differing from that vertex. Delete empty pieces and preserve earlier singletons. Remaining rectangles have $a \geq \tau$ and side at most ℓ. Equation (13), nonnegative omitted edges and normalized-mean orthogonality give

$$c\|U\|^2 \leq \mathcal{E}_a(U) + c\|C_{\mathcal{P}}U\|^2, \quad (C_{\mathcal{P}}U)_{R,j} = \sqrt{|R|}\,\mathrm{mean}_R\, U_j \tag{14}$$

for ALL cochains. Singleton residuals vanish, including $n = 1$; parallel edge occurrences for $n = 2$ are counted separately, and wrap edges can be omitted positively. With $B_c = (1 + \sqrt{c})^D$,

$$|\mathcal{P}| \leq B_c \max\{1, n^D/\tau^p\} + 2D\#\{a < \tau\}, \quad \#\{a < \tau\} \leq Mn^D\tau^p/m^p.$$

The singleton branch satisfies this count since $B_c\tau^{-p} > 1$ there. Taking $\tau = \sqrt{m}$ yields the weaker general bound

$$\dim F \leq h(B_c + 2DM)\max\{1, n^D/m^{D/4}\}.$$

6.2 Every global threshold can lose the target order

Let $\gamma = \max\{1, c/4\}$, $n = t^2$, $t \geq 2$, $m = \gamma n^2$, $b = \gamma n$, $S = \{0, \ldots, t-1\}^D$, and $f = n^{-p}$. Set $a = b$ on S, $a = A = (m - fb)/(1 - f)$ outside. Then $1 \leq a \leq 2m$, the mean is m, and the critical moment is less than $2 - f$. The target maximum is one. For any global $\tau \geq 1$, consider the preceding algorithm with any translated initial grid or excision order, retaining grid boundaries and every excised singleton. If $\tau \leq b$, no vertex is weak and the grid volume bound gives

$$|\mathcal{P}| \geq n^D/(2\sqrt{b/c})^D = (c/(4\gamma))^p n^{D/2},$$

or the singleton branch is larger. For large n the grid side is less than n. If $\tau > b$, all $t^D = n^{D/2}$ weak vertices remain distinct singletons. Every such threshold therefore loses the target order.

For the SAME profile retain the whole weak cube and split its complement by the first coordinate outside it, giving $D + 1$ rectangles. On S, $ct^2/(4b) = c/(4\gamma) \leq 1$; outside, $A \geq m$ and $cn^2/(4A) \leq 1$. Equation (13) proves (14) for this constant-size partition. This is a mechanism obstruction, not a counterexample to every rectangular partition.

6.3 Supplied scaled cubes attain the target order

Suppose additionally that disjoint integer cubes S_k of side L_k, nonwrapping in one common root, satisfy for fixed $\beta \geq 1$

$$\begin{aligned} cL_k(L_k - 1)/4 \leq a(x) \leq \beta L_k^2 \quad & (x \in S_k), \\ a(x) \geq cn(n-1)/4 \quad & (x \notin \cup_k S_k). \end{aligned} \tag{15}$$

These are extra coefficient and geometric assumptions. Each cluster charges at least β^{-p} to $\sum_x a(x)^{-p}$; thus $J \leq \beta^p M n^D/m^p$.

The solid-fat-object BSP construction of de Berg [1] has linear size in fixed dimension. Its first stage uses axis-parallel cuts; its second uses object facet planes. For solid axis-aligned cubes both stages are axis-parallel. Replace $S_k = \prod_i \{b_{ki}, \ldots, b_{ki} + L_k - 1\}$ by $Q_k = \prod_i [b_{ki} - 1/4, b_{ki} + L_k - 3/4]$.

These are positive-side solid cubes of fixed fatness, strictly disjoint even for touching integer-cell interpretations; their integer points are precisely S_k and are interior, including when $L_k = 1$.

Clip BSP leaves to $[-1/2, n-1/2]^D$, assign each cut boundary consistently to one child, take integer points and delete empty cells. If an assigned integer point belongs to S_k, a ball about it lies in Q_k, so the open leaf also meets Q_k. Hence one leaf's integer points belong to at most one cluster. Refine it with that cube's $2D$ facets, at most 3^D subrectangles, each inside one cluster or outside all clusters. Fragmentation is permitted; whole-cluster preservation is unnecessary. Consequently, for dimension-only $A_D \geq 1$,

$$|\mathcal{P}| \leq A_D \max\{1, J\}, \qquad \dim F \leq hA_D(1 + 2^p \beta^p M) \max\{1, n^D/(1+m)^p\}.$$

Inside a cluster fragment maximal side is at most L_k, outside at most n. Equations (15) and (13) prove (14). Projection separation and a high-budget restriction are unnecessary. This uses classical geometry; it does not derive the supplied cubes from the original moment.

6.4 The exact cube decomposition is not universal

For fixed c, β take arbitrarily large integer $n \geq 3$, $m = n$, $W = \{e_1, -e_1\}$, $f = 2/n^D$, with $a = 1$ on W and $a = A = (n-f)/(1-f)$ elsewhere. Then $n < A \leq 2n$, mean $a = n$, and the critical moment is less than $2n^{-p} + 1 - f \leq 2$. Choose also $n > \max\{\beta, 8\beta/c, 1 + 8/c\}$ and $n \geq c$. Every vertex has $a < cn(n-1)/4$, so any family satisfying (15) must cover zero. A side-one cube containing zero would require $A \leq \beta < n$. A cube of side $L \geq 2$ contains at least one of $e_1, -e_1$, in every translated root. Its lower bound forces $L^2 \leq 8/c$, while its upper bound at zero requires $A \leq \beta L^2 \leq 8\beta/c < n < A$. Even allowing overlap cannot repair this exact decomposition.

The SAME profile nevertheless admits a strong grid with $\ell = \min\{n, \lfloor 2\sqrt{A/c} \rfloor\}$ and two singleton excisions. Equation (14) holds and

$$\dim F \leq h[2^p(1 + \sqrt{c})^D + 4D] \max\{1, n^D/(1+m)^p\}.$$

There are at most $\lceil n/\ell \rceil^D + 4D$ blocks. If $\ell < n$, $\ell \geq \sqrt{A/c}$; if $\ell = n$, there is one grid box. Remaining rectangles have coefficient A and weak residuals are singletons. The universal supplied-cube route is refuted. The original unrestricted rectangular question is neither proved nor refuted.

7 Evidence provenance and verification limits

The analytic sources, their exact revisions and the executable finite diagnostics are collected in a hash-allowlisted companion package. Its map gives full SHAs and SHA256 values; short forms below are identifiers, not substitute hashes.

Item	Source revision	Verification status
Actual flat Hodge identity	`dfd49d0b0730`	earlier formal source
Critical spectral theorem	`527fcdcd119a`	analytic adapters
Dyadic spatial separation	`fb65a67af360`	analytic adapters
Higher moment block proof	`6dcb893155e5`	analytic adapters
Exact finite fixture code	`4bb460e1537e`	own Colab109
Full diagnostic report	`602ae3f0b59a`	finite rational batch

Colab109 checked four actual profiles on $(\mathbb{Z}/2\mathbb{Z})^4$ using rational arithmetic. Each normal and optimized Python mode accepted four profiles and rejected 16 real matrix, inverse or flow mutations,

with matching outputs and exit code zero. It checked the constant mode, forward symbols, actual pseudoinverse identities, feasible/minimizing flow energy, comparison inertia, strict low-spectrum multiplicities and noncommuting counts. There are 12 distinct density examples, evaluated again for each profile: 48 evaluations per mode, not 48 independent examples.

All eight files in the diagnostic archive's inner manifest were rehashed. Runtime execution was recorded at 17:13:06.300923–17:13:16.201862 UTC on 2 November2026, with automatic runtime release and zero sessions verified afterwards. Rounded resource units do not prove zero cost. No local Windows mathematical execution occurred. This single embedded-SHA finite batch is not two fresh matching clones, a Lean proof, a theorem about arbitrary dimensions or odd lattices, or independent mathematical review. No new Lean run occurs in this appendix's PDF assembly. Operational repository checks on unchanged Lean files do not verify these new analytic proofs.

There is also subsequent, strictly focal formal evidence. Source `41f8492c6190` supplies the ten four-cycle theorems and their oracle. Own Colab118 compiled that same source in two fresh clones; both actual oracle logs and compiled module hashes agree. Each of four normal/optimized validator runs accepted ten readouts and rejected three real corruptions. All readouts use only `propext`, `Classical.choice`, and `Quot.sound`. The full archive has 69 manifest files and is included with the failed Colab116/117 compiler attempts. Source hashes, complete logs and exact reports are allowlisted. This verifies Proposition 2's finite identities, not the geometric, shorted-operator, free-penalty or general rectangular adapters. Own Colab115 separately checked 18 finite diagnostics and eight real mutations in each mode of two fresh clones. Finite examples do not establish their general claims. Section 6 consolidates analytic sources at `0fa0d91000f6`, `312f476ea6e5`, `f52ae344f61f`, `1850be2aa6d5` and `b348f4488b3a`. These new spatial proofs have no Lean certificate or new numerical verification. PDF assembly does not change that status. The earlier exact PDF and its assessment remain preserved.

8 Model assessments and review context

The prior analytic-source first self-assessments were 6/10 for the critical spectral note, 5/10 for the dyadic note, and 5/10 for the exact finite diagnostic package. They apply only to their own exact source versions. All were by the preparing Codex agent; the served backend identity and effort were unexposed. They are neither independent model reviews nor human peer review. No earlier score is inherited by this appendix. The previous appendix PDF and its first exact assessment remain historical artifacts; their assessment does not transfer to this consolidated revision. This revision's first exact-PDF review is a separate SHA-linked sidecar after visual inspection. No grade-seeking repeated review, editorial acceptance or publication status is implied.

The principal limitations and objections are these. The new finite Fourier, inverse, field and partition adapters are analytic rather than kernel-verified. The classical Rumin–Frank method and elementary refinement geometry limit any novelty claim; a complete priority comparison is unfinished. The sufficient theorem uses one shared scalar coefficient profile, and its distribution condition is not proved from arithmetic means. Spectral coordinates are global; unrestricted critical rectangular means remain unresolved. Exact finite diagnostics cover a small selected family, and some count bounds are loose. The constants are safe rather than sharp. This appendix makes no statement about an interacting or continuum physical mass gap. PR29 remains 20/41 and TermSource=0.

9 The single remaining bounded question

For fixed $D \geq 3, c > 0, N_c \geq 2, M$, do all actual scalar profiles $1 \leq a \leq 2m$, exact mean m, satisfying (3), admit (4) using normalized means on an unrestricted axis-aligned integer rectangular partition? The mechanism must leave full cube trees, for example through anisotropic cuts with a proved variance bound. A falsifier must defeat every such rectangular partition, not merely every dyadic cube tree. After three substantive units without a proved reduction or useful falsifier, the route must be changed or its concrete blocker identified. Formalization and consolidation of the existing results are preserved work, not additional active new research questions.

References

[1] M. de Berg, *Linear Size Binary Space Partitions for Uncluttered Scenes*, UU-CS-1998-12 (1998), Section2.1, Lemmas2.1 and2.4, Theorem2.5 and Corollary2.9. `https://ics-archive.science.uu.nl/research/techreps/repo/CS-1998/1998-12.pdf`.

[2] W. N. Anderson, Jr. and G. E. Trapp, *Shorted Operators. II*, SIAM J. Appl. Math. 28(1) (1975), 60–71. `https://doi.org/10.1137/0128007`.

[3] R. L. Frank, *Cwikel's theorem and the CLR inequality*, arXiv:1206.3325v1 (2012), Section2, Theorem2.1, Corollary2.2, Lemmas2.4–2.5. `https://arxiv.org/abs/1206.3325`.

[4] S. Andres, J.-D. Deuschel and M. Slowik, *Harnack inequalities on weighted graphs and some applications to the random conductance model*, arXiv:1312.5473v6 (2019), Section2.2, equation(2.5) and Proposition2.1. `https://arxiv.org/abs/1312.5473`.